

Project FORESIGHT

AI-Powered Demand & Inventory Intelligence Platform

Notebook 04: Baseline Forecasting

Project Information

Item	Details
Author	Mohit Kumar
Internship	Zidio Development
Project Type	End-to-End Data Analytics & Machine Learning Project
Notebook Version	1.0
Status	Completed
Last Updated	August 2026

1. Business Context and Notebook Objective

A forecasting system needs a transparent and honestly evaluated benchmark before more complex models can be trusted. Machine-learning models developed later in Project FORESIGHT must demonstrate a reliable improvement over this benchmark rather than being selected only because they are more advanced.

Notebook 04 uses the validated weekly modelling dataset created in Notebook 03 to develop and compare statistical baseline forecasts for the approved 300 SKU-store series.

This notebook will:

- establish a leakage-safe chronological evaluation design;
- implement naive, seasonal-naive, moving-average and exponential-smoothing baselines;
- compare all candidates using identical validation periods and series coverage;
- select the champion baseline using validation evidence only;
- evaluate the champion once on the untouched final-test period;
- export the benchmark forecasts, metrics and configuration required by Notebook 05.

Scope boundary: This notebook covers baseline forecasting only. Machine-learning modelling belongs to Notebook 05, final baseline-versus-ML selection belongs to Notebook 06, and inventory decisioning belongs to Notebook 07.

⚙️ 2. Setup and Notebook 03 Handoff

This section prepares the forecasting environment and loads the approved Notebook 03 output. Only the minimum checks required for reliable baseline forecasting will be performed.

2.1 Environment, Paths and Input Loading

The required forecasting, data-handling and visualisation libraries will be imported, reproducibility settings will be established, and project paths will be defined.

The final weekly modelling dataset, `weekly_features_final.csv`, and only the supporting metadata required by this notebook will then be loaded. No data cleaning, EDA or feature reconstruction will be repeated.

Environment setup completed successfully.
Random seed: 42

Library Summary

The required Python libraries were successfully imported and configured for Notebook 04. Each library supports a specific part of the baseline forecasting workflow.

Library	Purpose
pandas	Loading, structuring, aggregating and validating forecasting data
NumPy	Numerical calculations, array operations and reproducibility controls
Matplotlib	Creation of forecasting diagnostics and performance charts
Seaborn	Statistical visualisation and consistent chart styling
Pathlib	Reproducible project-directory and file-path management
JSON	Reading and exporting champion-model configuration metadata
IPython.display	Enhanced display of tables and notebook outputs
statsmodels	Simple and damped-trend exponential-smoothing baselines

The random seed, display settings, warning controls and visualisation theme have also been configured consistently.

The notebook environment is now prepared for modelling-data loading and baseline forecasting.

Project root: Project FORESIGHT
Modelling data directory: data/modeling
Forecasting input: data/modeling/weekly_features_final.csv
Input file available: True

Forecasting dataset loaded successfully.
Dataset shape: 66,300 rows x 44 columns
Series column: sku_id
Weekly time column: week_start_date
Forecasting target: weekly_units_sold
Date columns parsed: {'week_start_date': 'datetime64[us]', 'week_end_date': 'datetime64[us]'}

	sku_id	wm_yr_wk	week_start_date	week_end_date	weekly_units_sold
0	FOODS_1_001_CA_1	11201	2012-01-28	2012-02-03	7
1	FOODS_1_001_CA_1	11202	2012-02-04	2012-02-10	5
2	FOODS_1_001_CA_1	11203	2012-02-11	2012-02-17	6



Setup and Input Summary

The notebook environment, reusable project paths and approved Notebook 03 modelling handoff were initialized successfully. The weekly dataset loaded under its verified schema and is ready for time-based validation.

This step confirms input accessibility and schema compatibility; complete forecasting readiness is evaluated separately in Section 2.2.

2.2 Minimal Input Readiness Check

A compact downstream-readiness check will confirm that the forecasting target, series identifiers and weekly time field are available and valid.

The check will also confirm 300 SKU-store series, unique SKU-store-week records, chronological ordering, usable target values and the weekly continuity required by the baseline methods.

This is a forecasting safety gate, not a repeat of the comprehensive validation and leakage controls completed in Notebooks 01 and 03.

	Check	Observed Result	Required Result	Status
0	Approved SKU-store series coverage	300 series	300 series	PASS
1	Missing values in model key	0 values	0 values	PASS
2	Duplicate SKU-store-week keys	0 rows	0 rows	PASS
3	Numeric forecasting target	True	True	PASS
4	Missing or non-finite target values	0 values	0 values	PASS
5	Negative target values	0 values	0 values	PASS

Core input-readiness checks passed successfully.

	Check	Observed Result	Required Result	Status
0	Missing weekly-date values	0 values	0 values	PASS
1	Non-chronological records within series	0 rows	0 rows	PASS
2	Duplicate SKU-store-date records	0 rows	0 rows	PASS
3	Non-seven-day intervals	0 intervals	0 intervals	PASS
4	Invalid weekly date spans	0 rows	0 rows	PASS
5	Equal observations across series	221 weeks per series	Same count for every series	PASS
6	Aligned date window across series	1 start date; 1 end date	1 common start and end date	PASS

Chronology and weekly-continuity checks passed successfully.

Input Readiness Summary

Together, the core and time-based validations confirm that every modelling record belongs to a reliable, chronologically ordered weekly series. The handoff can therefore support leakage-safe forecasting and evaluation without additional structural correction.

Section 2 Final Summary

The verified modelling handoff is ready for forecasting. Section 3 now defines the leakage-safe validation and final-test design.

3. Forecasting and Evaluation Design

This section establishes a consistent, leakage-safe framework for comparing all forecasting candidates before model performance is assessed.

3.1 Forecast Horizon and Chronological Splits

The task is to forecast weekly units sold for every SKU-store series across an eight-week horizon.

Model comparison will use three rolling-origin validation folds with expanding training windows. The latest eight weeks will remain an untouched final test and will not influence model comparison or champion selection.

At each origin, the model will be trained only on the history available at that time and will forecast the next eight weeks, reproducing the sequence required in real-world use.

	Design Component	Configured Value
0	Available panel history	221 weeks
1	Forecast horizon	8 weeks
2	Validation folds	3 folds
3	Final-test horizon	8 weeks
4	Reserved evaluation window	32 weeks
5	First validation training window	189 weeks
6	Seasonal reference period	52 weeks

	Split	Purpose	Training Weeks	Train Through	Horizon	Evaluation Window
0	Fold 1	Validation	189	2015-09-05	8	2015-09-12 to 2015-10-31
1	Fold 2	Validation	197	2015-10-31	8	2015-11-07 to 2015-12-26
2	Fold 3	Validation	205	2015-12-26	8	2016-01-02 to 2016-02-20
3	Final Test	Final Test	213	2016-02-20	8	2016-02-27 to 2016-04-16

	Check	Observed Result	Required Result	Status
0	Required split composition	3 validation; 1 final test	3 validation; 1 final test	PASS
1	Uniform forecast horizons	0 invalid splits	0 invalid splits	PASS
2	Consecutive train-evaluation boundaries	0 invalid boundaries	0 invalid boundaries	PASS
3	Eight-week training expansion	0 invalid expansions	0 invalid expansions	PASS
4	Non-overlapping evaluation windows	0 overlapping weeks	0 overlapping weeks	PASS
5	Latest continuous period reserved	Latest 32 weeks	Latest 32 weeks	PASS
6	Complete SKU-store coverage	300 series in every week	300 series in every week	PASS

Chronological Split Summary

The validated schedule preserves temporal order, consistent evaluation coverage and complete separation between training and forecast periods. All candidates can therefore be compared fairly, while the latest period remains protected for unbiased final testing.

3.2 Metrics and Forecasting Rules

Validation WAPE will serve as the primary model-selection metric. Forecast Bias, MAE and RMSE will provide supporting evidence on directional error, overall error magnitude and sensitivity to larger misses.

All candidates must use identical forecast origins, horizon lengths and SKU-store coverage. Predictions must remain non-negative, and every feature must be available at the corresponding forecast origin.

When future lag-based inputs depend on unknown demand, preceding predictions will be fed back recursively. Random data splitting, use of actual demand from within a forecast horizon, and any tuning or selection based on the final-test period are prohibited.

	Metric	Role	Definition	Preferred Result	Interpretation
0	WAPE	Primary	$\frac{\sum \text{Forecast} - \text{Actual} }{\sum \text{Actual}} \times 100$	Lower	Overall error relative to demand
1	Forecast Bias	Supporting	$\frac{\sum (\text{Forecast} - \text{Actual})}{\sum \text{Actual}} \times 100$	Closer to zero	Positive: over-forecast; negative: under-forecast
2	MAE	Supporting	Mean absolute error	Lower	Average error magnitude in units
3	RMSE	Supporting	Root mean squared error	Lower	Error measure emphasizing larger misses

	Control	Locked Setting	Notebook Enforcement	Status
0	Forecast range	Non-negative	Clip every prediction at zero	LOCKED
1	Multi-step generation	Recursive	Feed earlier predictions into later lag features	LOCKED
2	Within-horizon actual demand	Not permitted	Update history with predictions, not evaluatio...	LOCKED
3	Data partitioning	Chronological only	Reuse the approved splits without shuffling	LOCKED
4	Final-test use	Evaluation only	Evaluate only after champion selection	LOCKED

Evaluation Framework Summary

The approved framework evaluates accuracy, directional error and large misses under realistic forecasting conditions. Its common controls ensure that every candidate is assessed consistently without access to unavailable future information.

Section 3 Final Summary

The evaluation design is now fixed and ready to support consistent baseline development and model comparison in Section 4.



4. Baseline Models and Rolling-Origin Validation

This section implements the baseline forecasting candidates under the fixed rolling-origin design and establishes the benchmark for later model comparison.

4.1 Naive and Seasonal-Naive Baselines

The last-value naive baseline will extend the most recently observed weekly demand across the full forecast horizon. It will serve as the explicit reference benchmark for relative-improvement calculations.

The 52-week seasonal-naive baseline will forecast each horizon from the corresponding week of the preceding annual cycle. Because it preserves recurring retail seasonality without fitted parameters, it will provide a complementary seasonal comparator within the candidate set.

	Split	Minimum History Weeks	Complete Evaluation Series	Missing Seasonal References	Status
0	validation_fold_1	189	300	0	READY
1	validation_fold_2	197	300	0	READY
2	validation_fold_3	205	300	0	READY

	Candidate	Fold	Horizon	SKU-Store	Forecast Week	Actual	Forecast
0	Last-Value Naive	Fold 1	1	FOODS_1_001_CA_1	2015-09-12	3	5
1	Last-Value Naive	Fold 2	1	FOODS_1_001_CA_1	2015-11-07	4	3
2	Last-Value Naive	Fold 3	1	FOODS_1_001_CA_1	2016-01-02	4	3
3	52-Week Seasonal Naive	Fold 1	1	FOODS_1_001_CA_1	2015-09-12	3	5
4	52-Week Seasonal Naive	Fold 2	1	FOODS_1_001_CA_1	2015-11-07	4	4
5	52-Week Seasonal Naive	Fold 3	1	FOODS_1_001_CA_1	2016-01-02	4	4

	Check	Observed Result	Required Result	Status
0	Expected forecast volume	14,400 records	14,400 records	PASS
1	Unique forecast keys	0 duplicates	0 duplicates	PASS
2	Complete candidate-fold-horizon coverage	0 incomplete groups	0 incomplete groups	PASS
3	Finite actual and forecast values	0 invalid records	0 invalid records	PASS
4	Non-negative forecasts	0 negative forecasts	0 negative forecasts	PASS
5	Last-value calculation	0 mismatches	0 mismatches	PASS
6	Seasonal-naive calculation	0 mismatches	0 mismatches	PASS

Naive Benchmark Summary

Both transparent benchmarks now produce reliable validation forecasts under the approved design. Their verified outputs establish the reference point for assessing whether smoothing or fitted statistical methods provide dependable improvement.

4.2 Moving-Average Baselines

Moving averages based on the preceding 4, 8 and 13 weeks will represent short-, intermediate- and longer-term demand smoothing.

Each window will be updated recursively, allowing earlier forecasts to replace unavailable actual demand as the horizon advances.

	horizon	week_start_date	MA 13-Week	MA 4-Week	MA 8-Week
0	1	2015-09-12	3.54	4.00	3.88
1	2	2015-09-19	3.50	3.50	3.61
2	3	2015-09-26	3.69	3.38	3.81
3	4	2015-10-03	3.75	3.97	4.16
4	5	2015-10-10	3.88	3.71	3.93
5	6	2015-10-17	3.80	3.64	3.67
6	7	2015-10-24	3.63	3.67	3.63
7	8	2015-10-31	3.75	3.75	3.96

	Check	Observed Result	Required Result	Status
0	Expected forecast volume	21,600 records	21,600 records	PASS
1	Unique forecast keys	0 duplicates	0 duplicates	PASS
2	Complete candidate-fold-horizon coverage	0 incomplete groups	0 incomplete groups	PASS
3	Horizon and date alignment	0 mismatches	0 mismatches	PASS
4	Finite actual and forecast values	0 invalid records	0 invalid records	PASS
5	Non-negative forecasts	0 negative forecasts	0 negative forecasts	PASS
6	Actual-demand alignment	0 mismatches	0 mismatches	PASS
7	Recursive moving-average calculation	0 mismatches	0 mismatches	PASS

Moving-Average Summary

The recursive moving-average candidates now provide verified smoothing alternatives across different demand-memory lengths. Their validated forecasts are ready for later comparison with the transparent and fitted statistical baselines.

4.3 Exponential-Smoothing Baselines

Simple exponential smoothing and damped-trend exponential smoothing will be fitted independently to each evaluated SKU-store series.

Controlled fitting safeguards, documented fallback forecasts and non-negative output handling will preserve complete series coverage when isolated fitting problems occur.

	Candidate	Attempts	Fallbacks	Warning Fits	Clipped
0	Simple ES	900	0	0	0
1	Damped-Trend ES	900	0	0	129

	Check	Observed Result	Required Result	Status
0	Expected forecast volume	14,400 records	14,400 records	PASS
1	Expected fit-audit volume	1,800 records	1,800 records	PASS
2	Unique forecast keys	0 duplicates	0 duplicates	PASS
3	Unique fit-audit keys	0 duplicates	0 duplicates	PASS
4	Complete candidate-fold-horizon coverage	0 incomplete groups	0 incomplete groups	PASS
5	Horizon and date alignment	0 mismatches	0 mismatches	PASS
6	Forecast-origin alignment	0 mismatches	0 mismatches	PASS
7	Finite actual and forecast values	0 invalid records	0 invalid records	PASS
8	Non-negative forecasts	0 negative forecasts	0 negative forecasts	PASS
9	Actual-demand alignment	0 mismatches	0 mismatches	PASS
10	Fit-to-forecast path alignment	0 unmatched paths	0 unmatched paths	PASS
11	Fit-audit metadata consistency	0 invalid records	0 invalid records	PASS
12	Clipping traceability	0 mismatches	0 mismatches	PASS
13	Sampled model calculations	0 mismatches	0 mismatches	PASS

Exponential-Smoothing Summary

The fitted statistical baselines now produce complete and auditable multi-step forecasts under the approved safeguards. Their verified outputs are ready for comparative evaluation against the transparent and moving-average benchmarks.

4.4 Consolidated Validation Results

All validation forecasts will be assembled into a traceable record containing the candidate, fold, forecast origin, horizon, SKU-store identifiers, actual demand and predicted demand.

Performance will be summarized overall and across folds and horizons, with stability and coverage verified separately. SKU-store-level metrics will be prepared as validated handoff artefacts in Section 6.1.

Each candidate will then be assessed relative to the last-value naive benchmark to determine whether added complexity produces a dependable improvement.

	Candidate	Records	Folds	Horizons	Series	Status
0	Last-Value Naive	7200	3	8	300	READY
1	Seasonal Naive	7200	3	8	300	READY
2	MA (4-Week)	7200	3	8	300	READY
3	MA (8-Week)	7200	3	8	300	READY
4	MA (13-Week)	7200	3	8	300	READY
5	Simple ES	7200	3	8	300	READY
6	Damped-Trend ES	7200	3	8	300	READY

	Candidate	Method Family	WAPE (%)	MAE	RMSE	Bias (%)
0	damped_trend_exponential_smoothing	Exponential smoothing	50.02	3.75	7.37	0.21
1	simple_exponential_smoothing	Exponential smoothing	50.10	3.75	7.39	-0.07
2	moving_average_8_week	Moving average	50.43	3.78	7.32	0.70
3	moving_average_4_week	Moving average	50.92	3.81	7.34	1.04
4	moving_average_13_week	Moving average	51.54	3.86	7.68	1.81
5	naive_last_value	Transparent benchmark	58.70	4.39	8.79	-0.35
6	seasonal_naive_52_week	Transparent benchmark	69.64	5.21	10.21	-7.26

	Candidate	Mean Fold WAPE (%)	WAPE Std. Dev.	WAPE Range	Worst Fold WAPE (%)	Mean Fold Rank	Fold Wins
0	simple_exponential_smoothing	50.08	3.11	6.73	54.48	2.00	1
1	damped_trend_exponential_smoothing	49.99	3.20	6.83	54.52	2.33	1
2	moving_average_8_week	50.41	3.08	7.37	54.55	2.67	1
3	moving_average_4_week	50.89	3.76	8.72	56.10	3.33	0
4	moving_average_13_week	51.52	2.51	6.15	54.68	4.67	0
5	naive_last_value	58.67	3.77	8.55	63.95	6.00	0
6	seasonal_naive_52_week	69.65	1.72	4.20	71.82	7.00	0

	Candidate	H1 WAPE	H8 WAPE	Change	Worst WAPE	Mean Rank	Wins
0	Damped-Trend ES	43.56	58.90	15.34	58.90	2.00	4
1	Simple ES	43.54	58.76	15.22	58.76	2.50	1
2	MA (8-Week)	44.82	58.56	13.74	58.56	2.88	1
3	MA (4-Week)	44.39	59.86	15.47	59.86	3.62	0
4	MA (13-Week)	46.71	58.42	11.71	58.42	4.00	2
5	Last-Value Naive	51.76	67.75	15.99	67.75	6.00	0
6	Seasonal Naive	67.24	69.90	2.66	79.81	7.00	0

	Candidate	WAPE (%)	Bias (%)	Fold Rank	Horizon Rank	vs Naive (%)	Decision
0	Damped-Trend ES	50.02	0.21	2.33	2.00	14.79	CHAMPION
1	Simple ES	50.10	0.07	2.00	2.50	14.64	FINALIST
2	MA (8-Week)	50.43	0.70	2.67	2.88	14.08	FINALIST
3	MA (4-Week)	50.92	1.04	3.33	3.62	13.25	BENCHMARK
4	MA (13-Week)	51.54	1.81	4.67	4.00	12.19	BENCHMARK
5	Last-Value Naive	58.70	0.35	6.00	6.00	0.00	BENCHMARK
6	Seasonal Naive	69.64	7.26	7.00	7.00	-18.65	BENCHMARK

Baseline-Selection Summary

The damped-trend exponential-smoothing baseline was selected as the champion because it achieved the strongest overall validation WAPE while remaining consistently competitive across validation folds and forecast horizons. Simple exponential smoothing and the eight-week moving average remained credible finalists, but neither surpassed the champion on the approved primary selection metric. The selected baseline is now locked for one-time evaluation on the untouched final-test period.

Section 4 Completion

The baseline-development stage is complete. Seven forecasting candidates were generated and validated through rolling-origin evaluation, followed by overall, fold-wise, and horizon-wise performance assessment. The damped-trend exponential-smoothing model has been selected and locked as the champion baseline, while the final-test period remains untouched for unbiased evaluation in the next section.

5. Champion Baseline and Final-Test Evaluation

This section carries the validation-selected champion forward under a locked configuration and evaluates its generalisation performance once on the untouched final-test period.

5.1 Lock Champion Configuration and Final-Test Boundary

The selected `damped_trend_exponential_smoothing` baseline will be locked together with its forecasting method, eight-week horizon and chronological final-test boundary.

Only configuration and split metadata will be verified at this stage. Final-test actual demand will remain unaccessed until the champion forecasts have been generated and locked.

	Locked Component	Locked Value
0	Champion baseline	<code>damped_trend_exponential_smoothing</code>
1	Model method	<code>damped_trend</code>
2	Validation WAPE	50.02%
3	Improvement vs naive	14.79%
4	Training end	2016-02-20
5	Final-test start	2016-02-27
6	Final-test end	2016-04-16
7	Forecast horizon	8 weeks
8	Series coverage	300 series
9	Configuration status	LOCKED

- ✓ Champion configuration and final-test boundary locked successfully.
- 🔒 Final-test actual demand remains untouched.

✓ Champion Lock Summary

The validation-selected `damped_trend_exponential_smoothing` baseline is now locked with an eight-week forecast horizon across all 300 SKU-store series.

The final training week ends on **20 February 2016**, followed immediately by the untouched final-test period from **27 February 2016 to 16 April 2016**. Final-test actual demand remains unaccessed.

5.2 Untouched Final-Test Evaluation

The locked champion will be refitted using all demand history available strictly before the final-test cutoff and used to forecast the untouched eight-week period.

Final-test actuals will then be accessed once to calculate WAPE, Forecast Bias and coverage using the same rules applied during validation. These results will provide a transparent generalisation assessment and will not be used for further tuning or model selection.

Forecast Component		Result
0	Champion baseline	damped_trend_exponential_smoothing
1	Forecast records	2,400
2	Series covered	300
3	Forecast horizon	8 weeks
4	First forecast week	2016-02-27
5	Last forecast week	2016-04-16
6	Model fits	300
7	Warnings recorded	0
8	Forecast status	LOCKED

- ☒ Champion final-test forecasts generated and locked successfully.
 Final-test actual demand remains untouched.

Scoring Component		Result
0	Forecast records	2,400
1	Actual records	2,400
2	Scored records	2,400
3	Series covered	300
4	Test weeks	8
5	Missing actuals	0
6	Missing predictions	0
7	Actual-data access	ACCESSED ONCE
8	Scored-dataset status	LOCKED

- ☒ Final-test actuals accessed once and matched to the locked forecasts successfully.
 Final-test scored dataset locked: 2,400 records.

Metric		Result
0	WAPE	53.17%
1	Forecast Bias	10.94%
2	MAE	3.2138
3	RMSE	5.9251
4	Forecast Coverage	100.00%
5	Total Actual Demand	14,506.00
6	Total Predicted Demand	16,093.14

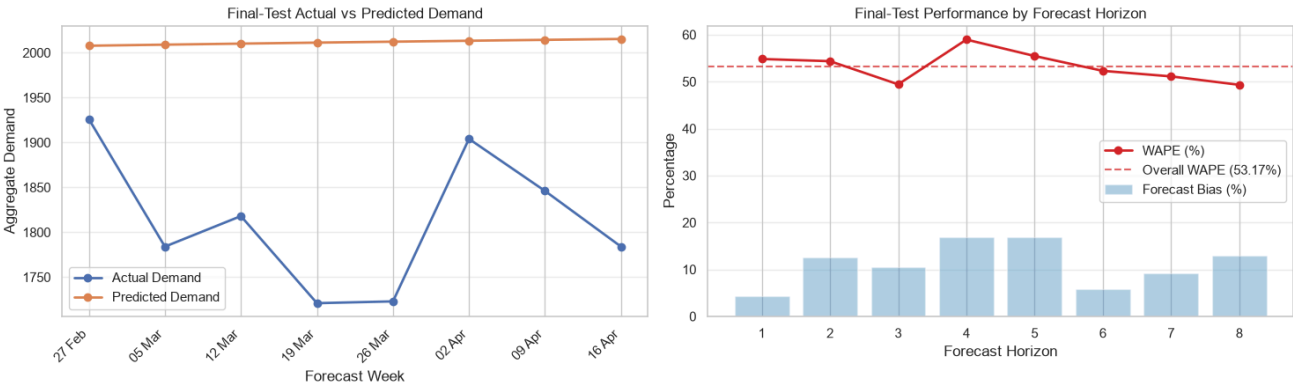
	Horizon	Forecast Week	Records	Actual Demand	Predicted Demand	WAPE (%)	Bias (%)	MAE	RMSE
0	1	2016-02-27	300	1926	2,007.75	54.82	4.24	3.52	7.90
1	2	2016-03-05	300	1784	2,008.95	54.36	12.61	3.23	6.06
2	3	2016-03-12	300	1818	2,010.09	49.42	10.57	3.00	4.85
3	4	2016-03-19	300	1721	2,011.20	58.95	16.86	3.38	6.05
4	5	2016-03-26	300	1723	2,012.27	55.46	16.79	3.19	5.80
5	6	2016-04-02	300	1904	2,013.30	52.29	5.74	3.32	5.63
6	7	2016-04-09	300	1846	2,014.30	51.11	9.12	3.15	5.78
7	8	2016-04-16	300	1784	2,015.28	49.32	12.96	2.93	4.78

	Metric	Validation (%)	Final Test (%)	Change (pp)	Assessment
0	WAPE	50.02	53.17	3.16	Degraded
1	Absolute Forecast Bias	0.21	10.94	10.73	Degraded

✓ Final-test overall, horizon-wise and generalisation metrics calculated successfully.

i Positive bias indicates over-forecasting; negative bias indicates under-forecasting.

Champion Baseline — Untouched Final-Test Diagnostics



	Validation Control	Status
0	Champion configuration remains locked	PASS
1	Final-test forecasts remain locked	PASS
2	Scored dataset remains locked	PASS
3	Expected evaluation records retained	PASS
4	Unique series-week scoring key	PASS
5	All approved series retained	PASS
6	All eight horizons retained	PASS
7	Locked test boundary retained	PASS
8	Forecast origin precedes test period	PASS
9	Actual-demand totals reconcile	PASS
10	Predicted-demand totals reconcile	PASS
11	Stored WAPE reconciles	PASS
12	Stored Forecast Bias reconciles	PASS
13	Complete forecast coverage retained	PASS

✓ Final-test evaluation validated: 14 of 14 controls passed.

🚩 Untouched final-test evaluation completed with WAPE 53.17% and Forecast Bias +10.94%.

✓ Final-Test Evaluation Summary

The champion achieved a final-test **WAPE of 53.17%**, compared with **50.02%** during validation—a deterioration of **3.16 percentage points**.

Rounding note: The deterioration of **3.16 percentage points** is calculated from the underlying unrounded WAPE values. Subtracting the displayed rounded values gives **3.15 percentage points**.

Forecast Bias increased from **+0.21%** during validation to **+10.94%** on the final test. Positive bias across all eight forecast horizons indicates a consistent tendency to over-forecast demand. Horizon-wise WAPE ranged from **49.32% to 58.95%**.

All **14 evaluation-integrity controls** passed. These results therefore form the unbiased baseline benchmark, and no further tuning is permitted using final-test evidence.

🚩 Section 5 Completion

Section 5 has established the approved baseline benchmark through validation-based champion selection and one-time holdout evaluation.

The next section will persist the resulting artefacts, verify their reproducibility and prepare the Notebook 05 handoff.

6. Outputs, Conclusion and Notebook 05 Handoff

This section converts the validated Notebook 04 results into reusable outputs and completes the baseline-modelling handoff to Notebook 05.

6.1 Export and Validate Outputs

The following machine-readable handoff artefacts will be exported to `data/processed/baseline_outputs` :

- `04_baseline_forecasts.csv`
- `04_baseline_model_metrics.csv`
- `04_baseline_sku_metrics.csv`
- `04_champion_baseline_config.json`

The following reporting tables will be saved to `reports/tables` :

- `04_baseline_selection_scorecard.csv`
- `04_baseline_validation_fold_metrics.csv`
- `04_baseline_validation_horizon_metrics.csv`
- `04_baseline_final_test_overall_metrics.csv`
- `04_baseline_final_test_horizon_metrics.csv`
- `04_baseline_final_test_validation_summary.csv`

The champion diagnostic figure will be saved as `reports/figures/04_baseline_champion_final_test_diagnostics.png` .

A reload-based validation will verify file availability, required schemas, unique forecast keys, series and horizon coverage, metric reconciliation, and consistency with the locked champion configuration.

	Prepared Artefact	Prepared Records	Status
0	04_baseline_forecasts.csv	52800	READY
1	04_baseline_model_metrics.csv	7	READY
2	04_baseline_sku_metrics.csv	2400	READY
3	04_champion_baseline_config.json	1	READY

✓ Baseline handoff datasets prepared and validated successfully.

📁 Baseline output location: `data/processed/baseline_outputs`

	Type	Artefact	Records	Status
0	Handoff Dataset	04_baseline_forecasts.csv	52800	WRITTEN
1	Handoff Dataset	04_baseline_model_metrics.csv	7	WRITTEN
2	Handoff Dataset	04_baseline_sku_metrics.csv	2400	WRITTEN
3	Configuration	04_champion_baseline_config.json	1	WRITTEN
4	Reporting Table	04_baseline_selection_scorecard.csv	7	WRITTEN
5	Reporting Table	04_baseline_validation_fold_metrics.csv	21	WRITTEN
6	Reporting Table	04_baseline_validation_horizon_metrics.csv	56	WRITTEN
7	Reporting Table	04_baseline_final_test_overall_metrics.csv	7	WRITTEN
8	Reporting Table	04_baseline_final_test_horizon_metrics.csv	8	WRITTEN
9	Reporting Table	04_baseline_final_test_validation_summary.csv	14	WRITTEN
10	Diagnostic Figure	04_baseline_champion_final_test_diagnostics.png	1	WRITTEN

✔ Notebook 04 exports completed: 11 of 11 files written successfully.

🧹 Obsolete Notebook 04 files removed: 0.

📦 Handoff artefacts: data/processed/baseline_outputs

📊 Reporting tables: reports/tables

🖼️ Diagnostic figure: reports/figures/04_baseline_champion_final_test_diagnostics.png

	Validation Control	Status
0	All 11 exported files are available	PASS
1	Forecast record count retained	PASS
2	Model metric record count retained	PASS
3	SKU-store metric record count retained	PASS
4	Forecast schema retained	PASS
5	Unique forecast keys retained	PASS
6	Forecast values remain complete	PASS
7	Evaluation stages retained	PASS
8	Validation forecast coverage retained	PASS
9	Final-test forecast coverage retained	PASS
10	Final-test series coverage retained	PASS
11	Final-test horizon coverage retained	PASS
12	Forecast chronology retained	PASS
13	Forecast dataset values reconcile	PASS
14	Model metric values reconcile	PASS
15	Unique SKU-store metric keys retained	PASS
16	SKU-store metric values reconcile	PASS
17	All reporting-table values reconcile	PASS
18	Champion configuration remains locked	PASS
19	Final-test completion status retained	PASS
20	Configuration file inventory retained	PASS
21	Stored final-test metrics reconcile	PASS
22	Diagnostic figure is readable	PASS

✓ Baseline outputs reloaded and validated: 23 of 23 controls passed.

📦 Notebook 04 baseline handoff is complete and ready for Notebook 05.

✓ Output Validation Summary

All **11 baseline outputs** were successfully reloaded, and all **23 validation controls** passed.

The saved artefacts faithfully preserve the validated forecasts, metrics, reporting outputs and locked champion configuration. Section 6.1 has therefore established a complete and reproducible baseline output package.

6.2 Key Outcomes and Notebook 05 Handoff

The `damped_trend_exponential_smoothing` model is the approved baseline champion. Two distinct benchmarks now govern subsequent modelling:

Benchmark	Result	Purpose
Rolling-validation WAPE	50.02%	Model development and candidate selection
Untouched final-test WAPE	53.17%	Reserved reference for later generalisation comparison

The baseline provides complete forecasts but remains limited by its univariate design. It cannot directly learn from price, promotion, calendar, event or hierarchy features; it also models each SKU-store series independently. Its **+10.94% final-test bias** further indicates sensitivity to demand changes outside the historical trend.

Notebook 05 must therefore:

- preserve the weekly SKU-store grain, approved 300-series scope and eight-week horizon;
- use the same chronological rolling-origin folds and metric definitions;
- calculate every feature using information available at the forecast origin;
- select machine-learning candidates using validation evidence only;
- target lower validation WAPE, lower absolute bias and stable performance across folds and horizons;
- and maintain complete forecast coverage.

The untouched final-test results must not be used for feature selection, hyperparameter tuning or machine-learning candidate selection.

🚩 Section 6 Completion

Section 6 has exported and validated the complete baseline output package and formalised the requirements for subsequent modelling.

✅ Notebook 04 Complete

Notebook 04 has established the approved, reproducible statistical baseline benchmark for Project FORESIGHT.

The project is now ready to proceed to **Notebook 05 — Machine-Learning Forecasting**.